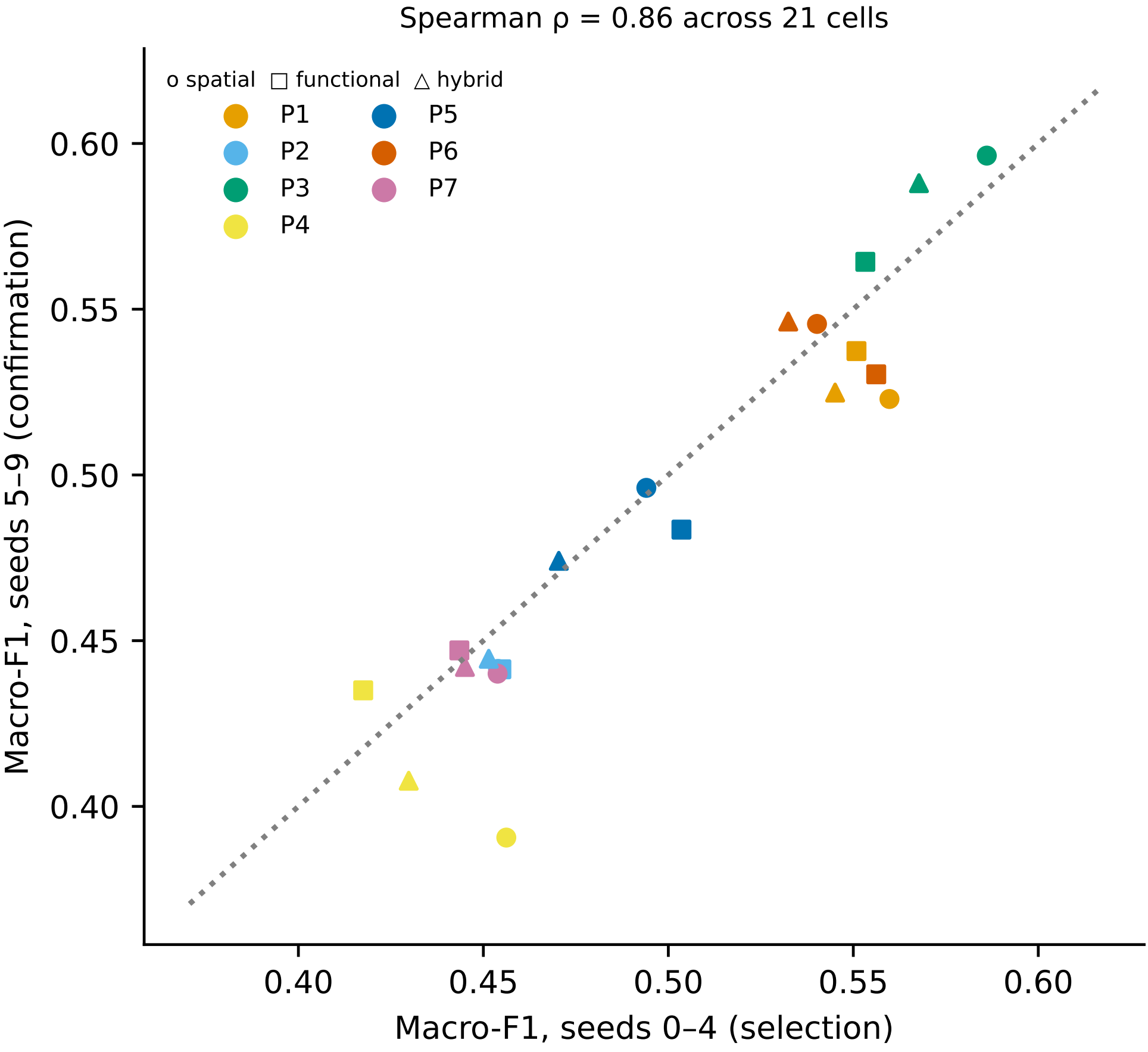
Spearman ρ = 0.86 across 21 cells
o spatial □ functional △ hybrid
P1
P2
P3
P4
P5
P6
P7
Macro-F1, seeds 5–9 (confirmation)
0.60
0.55
0.50
0.45
0.40
0.40
0.45
0.50
0.55
0.60
Macro-F1, seeds 0–4 (selection)